AIS — Lab reference guide

Key Concepts: Active Directory + GLPI

Site A project — Windows Server 2022 (AD DS, DNS) · Debian 13 · GLPI 11 · LDAP

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1. The big picture

A company needs two things for its IT users: **one identity per person** and **one place to ask for help**.

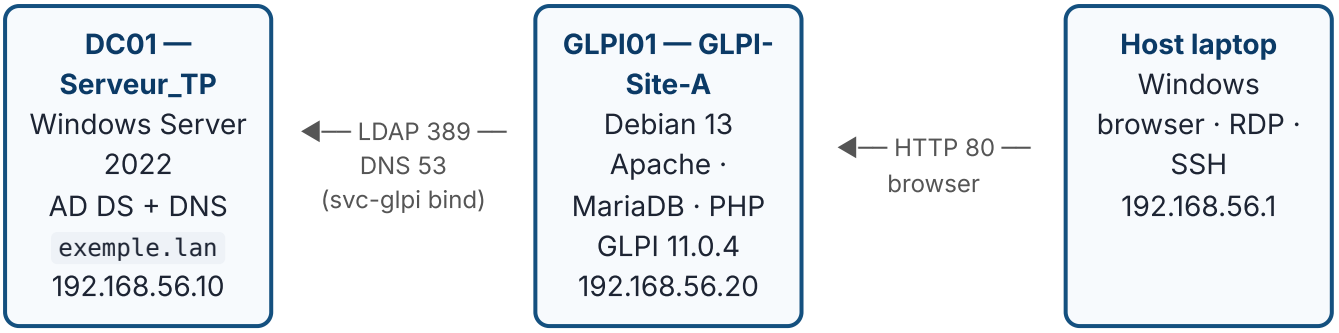
- **Active Directory (AD)** is the identity system. It stores who the users are, which groups they belong to, and checks their passwords.
- **GLPI** is the IT service-management tool: help-desk tickets plus the inventory of computers and network devices.
- **LDAP** is the language that lets GLPI ask AD "does this user exist and is this password right?". Users log in to GLPI with their normal Windows account; no second password is needed.

> **Analogy:** AD is the company's HR register and badge office. GLPI is the help desk at reception. LDAP is the internal phone line the receptionist uses to call HR and check a badge before helping someone.

The project follows a real "put into production" workflow:

1 Inventory → 2 Network & DNS → 3 AD structure → 4 LDAP link → 5 Profiles → 6 Asset inventory → 7 Ticket → 8 Acceptance tests

2. Lab architecture and IP plan



	Home lab	School design (target)
Server network	VirtualBox host-only 192.168.56.0/24 (flat)	VLAN 30 – Server 172.16.0.64/27
Network devices	None (declared manually in GLPI)	SW1 172.16.0.129, R1 172.16.0.162, FW 172.16.0.174
Internet for GLPI	Second adapter in NAT mode	Through the router and firewall

> **Rule to remember:** two machines talk directly only if they are on the **same subnet**. Otherwise a **router** is needed. At the start AD (192.168.20.x) and GLPI (192.168.56.x) were on different subnets with no router, so nothing could work until GLPI was moved to 192.168.56.20.

3. Networking basics that made or broke the lab

Concept	Plain meaning	Where it mattered
IP address / mask	The address of a machine and the size of its neighbourhood (/24 = 256 addresses, same first three numbers).	Putting AD and GLPI in 192.168.56.0/24.
Gateway	The "exit door" to other networks.	GLPI uses the NAT adapter's gateway for internet access; the host-only adapter has none.
VirtualBox **NAT**	The VM reaches the internet through the host, but nobody can reach the VM.	GLPI's internet access (apt install).
VirtualBox **Host-only**	A private network between the host and its VMs.	The lab network 192.168.56.0/24.
"Not attached" / cable connected	A virtual network card can exist with no cable plugged in.	The first fault found in Stage 1.
Snapshot	A restore point of a VM.	Taken before every change.
Full clone	An independent copy of a VM, with new MAC addresses.	GLPI-Site-A was cloned from debian-tp so another class's VM stayed untouched.

Port	A numbered door for a service on a machine.	22 SSH, 53 DNS, 80 HTTP, 389 LDAP, 636 LDAPS, 3389 RDP.

Debian network files

- `/etc/network/interfaces`: static IP configuration (ifupdown). Apply it with `ifdown/ifup <iface>`.
- `/etc/resolv.conf`: which DNS server to ask. It can be overwritten by DHCP clients; in this lab **dhcpcd** kept replacing it, which was fixed with `nohook resolv.conf` in `/etc/dhcpcd.conf`.
- `/etc/hosts` and `hostnamectl`: the machine's own name (glpi01).

4. DNS — the phone book of the domain

DNS (Domain Name System) turns names into IP addresses: `glpi01.exemple.lan → 192.168.56.20`. Active Directory **depends on DNS**. Clients find the domain controller through DNS records, so if DNS is wrong, AD looks "broken" even when it is fine.

Record	Meaning	Example
A	Name → IPv4 address	glpi01 → 192.168.56.20
PTR (reverse zone)	IP address → name	192.168.56.20 → glpi01.exemple.lan
SRV	"Where is the service?" (used to locate the DCs)	_ldap._tcp.exemple.lan
Zone	The set of records for one domain	exemple.lan

> **Lessons from the lab:**
> • A DC must use **itself** as its DNS server, not 8.8.8.8, or it cannot find its own domain.
> • An old network card left enabled had re-registered a stale address (192.168.20.10), so `nslookup` returned two answers. Disable the unused card, then delete the stale records.
> • Always test both ways: name → IP and IP → name.

5. Active Directory

Term	Plain meaning	In this project
Domain	A security boundary sharing one user database.	`exemple.lan`
AD DS	The Windows role that provides the directory.	Installed on Serveur_TP
DC (Domain Controller)	The server holding the directory and checking logins.	DC01 / WIN-RG5VDD2MS10
OU (Organizational Unit)	A folder for objects. It lets you apply a GPO or delegate admin rights to one part of the directory.	Utilisateurs (Direction, Employes, IT), Groupes, Postes, ComptesService

Security group (global)	A list of users who get the same rights. Grant rights to groups, never to individual users.	GRP-Direction, GRP-Employes, GRP-IT, GRP-GLPI-SVC
User account	A real person.	direction01, employe01, it01
Service account	An account used by an application rather than a person. Minimum rights, no interactive login.	svc-glpi (only reads the directory)
GPO	A Group Policy Object: settings pushed to the users and computers in an OU.	Reason for separating OUs

```
DC=exemple,DC=lan
├── OU=Utilisateurs
│   ├── OU=Direction        direction01  → GRP-Direction
│   ├── OU=Employes         employe01    → GRP-Employes
│   └── OU=IT               it01         → GRP-IT
├── OU=Groupes              GRP-*
├── OU=Postes               (computers)
└── OU=ComptesService       svc-glpi     → GRP-GLPI-SVC
```

> **Why separate OUs?** Different objects need different rules (GPOs, delegation, monitoring). Separate OUs also let GLPI search only `OU=Utilisateurs`, so service accounts are never imported as people.

6. LDAP — how GLPI talks to Active Directory

LDAP (Lightweight Directory Access Protocol) is the standard way to **search** a directory and **check a password** against it. AD answers LDAP on port **389** (unencrypted) and **636** (LDAPS, encrypted with TLS).

> **How a GLPI login works:**
> 1. GLPI connects to the DC as `svc-glpi` (the **bind**).
> 2. It **searches** under the Base DN for a user whose `sAMAccountName` = what was typed.
> 3. It re-binds as that user with the typed password. If AD accepts, the login succeeds.
> 4. It identifies the person by `objectGUID` so their tickets and assets stay linked to them.

GLPI setting	Meaning	Lab value
Server / Port	Where the DC listens	192.168.56.10 / 389
DN (Distinguished Name)	The full "address" of an object in the tree, read right to left	CN=svc-glpi,OU=ComptesService,DC=exemple,DC=lan
Base DN	Where in the tree to start searching	DC=exemple,DC=lan
Bind DN	The account GLPI uses to connect and search	svc-glpi (read only, not an admin)

Login field	The attribute matched against the typed username	samaccountname
Sync field	A stable unique ID	objectguid
Filter	Which objects count as "users"	`(&(objectClass=user)(objectCategory=person))`

Key attributes

- **sAMAccountName**: the short login name (it01). It can change.
- **objectGUID**: a unique ID that **never changes**, even after a rename or a move to another OU. That is why it is used for synchronisation.
- **CN / OU / DC** in a DN: Common Name / Organizational Unit / Domain Component.
- **Referrals**: AD can answer "ask another server". GLPI usually has referrals turned off for a single DC.

Testing LDAP outside GLPI (from Debian)

```
timeout 3 bash -c '</dev/tcp/192.168.56.10/389' && echo "Port 389 OPEN"
ldapsearch -x -H ldap://192.168.56.10 -D "CN=svc-
glpi,OU=ComptesService,DC=exemple,DC=lan" -W \
  -b "DC=exemple,DC=lan" "(sAMAccountName=it01)" dn
# -W prompts for the password: it is never written in the command or history
```

"result: 0 Success" plus the user's DN shows that network, DNS, the account and the Base DN are all correct, so any remaining problem is in the GLPI settings.

7. GLPI

GLPI (Gestionnaire Libre de Parc Informatique, "free IT asset manager") is a web application (PHP + MariaDB) for **ITSM**: help desk, asset inventory, contracts and knowledge base. In GLPI 11 the URL is the server root (`http://192.168.56.20/`), not `/glpi` .

Profiles (what a user may do)

Profile	For whom	Can do
Self-Service	Ordinary users (direction01, employe01)	Open and follow their own tickets, approve solutions. No configuration access.
Technician	IT staff (it01)	Handle tickets, view and edit assets. Cannot assign a ticket to a group (observed in the lab).
Super-Admin	Platform administrator	Everything, including configuration.

- **Authorizations** (Habilitations) = profile + entity (+ recursive). A user can have several.
- **Dynamic profile via rules**: at import, an authorization rule gave Self-Service automatically. Technician was added to it01 by hand.
- **GLPI groups** (e.g. IT) are used to assign tickets to a team. They are separate from AD groups unless synchronised.

Entities vs Locations — a classic confusion

	Entity	Location
What it is	An **administrative / visibility** boundary (who can see what)	A **physical** place (where the thing is)
Effect	Controls rights and data isolation	Information only, no effect on rights
Lab	Root > Site A > Direction, Employés, IT, Serveurs	Site A > Bureau Direction, Open space Employés, Local IT, Salle serveurs, Baie réseau

Inventory

- **GLPI Agent**: a small program installed on a computer that sends its hardware and software inventory to GLPI automatically (the server URL is set during install; a first inventory can be forced at `http://localhost:62354`).
- **Manual declaration**: used for machines without an agent and for network devices (switch, router, firewall).
- Each asset should be linked to a **user**, a **location** and a **technician / group** in charge.

8. ITIL ticket lifecycle

ITIL is a set of IT service-management best practices. GLPI follows its vocabulary.

New → Processing (assigned) → Pending (optional) → Solved → Closed

Term	Meaning
Incident	Something is broken or degraded ("I can't reach the intranet").
Request	Asking for something new ("I need software installed").
Requester / Assigned to / Observer	The ticket's actors: who asked, who works on it, who is kept informed.
Urgency × Impact → Priority	How fast the requester needs it × how many people are affected = the order of processing.
ITIL category	Classification (e.g. Network > Access to an internal service), used for routing and statistics. Managed at `/front/itilcategory.php`.
Follow-up / Task	A message in the conversation / a technical action with time spent.
Solved vs Closed	Solved = the technician says it is fixed and documents the solution. Closed = the requester approves (or a delay expires). Until it is closed, the requester can refuse the solution and the ticket reopens.
History	An automatic audit trail of every change: proof of the lifecycle.

> **Lab run:** employe01 created an incident (Network category, item CLIENT01) → the supervisor assigned it to it01 and the IT group → it01 added a diagnosis and a solution → Solved → employe01 approved → Closed.

9. Security principles applied

- **Least privilege**: svc-glpi can only read the directory; it is not a Domain Admin. If GLPI were compromised, the attacker would not own the domain.
- **Separation of duties**: the technician works the tickets, and a supervisor handles group assignment and configuration.
- **No secrets in documentation**: passwords are typed at prompts (`Read-Host -AsSecureString`, `ldapsearch -W`) and never captured in screenshots or reports.
- **Change safety**: a snapshot before every change; a full clone instead of modifying a VM used by another class.
- **Encryption**: LDAP on port 389 sends data in clear text; production should use **LDAPS (636)** with a certificate.
- **Root SSH login disabled**: log in as a normal user, then `su -`.

10. Troubleshooting method

Always go from the bottom layer up. Each step depends on the one before it:

1 Network (cable / adapter) → 2 IP / subnet → 3 DNS → 4 AD service → 5 LDAP (port + bind) → 6 GLPI settings

Symptom met	Cause	Fix
`Get-ADDomain` fails, interfaces down	VirtualBox adapter "not attached" / cable disconnected	Attach to host-only, tick "Cable connected"
AD and GLPI cannot ping each other	Different subnets, no router	Same subnet 192.168.56.0/24
`nslookup` returns two addresses	Old network card registering a stale A record	Disable the card, delete the old records
GLPI resolves nothing in exemple.lan (NXDOMAIN)	dhcpcd rewrote resolv.conf	`nohook resolv.conf`, restart the interface
GLPI "item not found"	Wrong URL (/glpi)	Use the server root
Copy/paste broken in RDP	Conflict between the VirtualBox shared clipboard and rdpclip	Disable the VirtualBox clipboard, sign out, reconnect
RDP session kicked out	The same user logged in at the console	One session per user: use RDP only
Root SSH "permission denied"	Root login disabled (good practice)	Log in as stagiaire, then `su -`

11. Command cheat sheet

Windows Server / PowerShell

```
ipconfig /all                                    # IP, DNS servers, adapters
Get-ADDomain                                     # is the domain reachable?
Get-ADOrganizationalUnit -Filter * | Select Name, DistinguishedName
New-ADOrganizationalUnit -Name "IT" -Path "OU=Utilisateurs,DC=exemple,DC=lan"
New-ADGroup -Name "GRP-IT" -GroupScope Global -GroupCategory Security -Path
"OU=Groupes,DC=exemple,DC=lan"
New-ADUser -Name it01 -SamAccountName it01 -Path
"OU=IT,OU=Utilisateurs,DC=exemple,DC=lan" `
  -AccountPassword (Read-Host -AsSecureString "Password") -Enabled $true
Add-ADGroupMember -Identity "GRP-IT" -Members it01
Get-ADPrincipalGroupMembership svc-glpi | Select Name   # check it is NOT an admin
Get-DnsServerResourceRecord -ZoneName exemple.lan -RRType A
Add-DnsServerResourceRecordA -ZoneName exemple.lan -Name glpi01 -IPv4Address
192.168.56.20 -CreatePtr
Set-DnsClientServerAddress -InterfaceAlias "Ethernet" -ServerAddresses 127.0.0.1
Disable-NetAdapter -Name "Ethernet 2"
Resolve-DnsName glpi01.exemple.lan
```

Debian

```
ip a ; ip r                                      # addresses and routes
cat /etc/network/interfaces ; cat /etc/resolv.conf
ifdown enp0s8 && ifup enp0s8                     # re-apply interface config
hostnamectl set-hostname glpi01
ping -c 3 192.168.56.10 ; nslookup dc01.exemple.lan
systemctl status apache2 mariadb ssh
php -m | grep -i ldap                            # PHP LDAP module present?
apt install ldap-utils                           # provides ldapsearch
```

VirtualBox (host)

```
VBoxManage list vms ; VBoxManage list runningvms
VBoxManage showvminfo "Serveur_TP" | findstr NIC
VBoxManage snapshot "Serveur_TP" take "before-stage3"
```

12. Glossary of acronyms

Acronym	Stands for	In one line
AD / AD DS	Active Directory (Domain Services)	Microsoft's directory of users, groups and computers
AIS	Administrateur d'Infrastructures Sécurisées	The French professional qualification this lab belongs to
CN	Common Name	The name part of a DN (CN=svc-glpi)
DC	Domain Controller / Domain Component	The AD server / a part of the domain name in a DN (DC=lan)

DHCP	Dynamic Host Configuration Protocol	Hands out IP settings automatically
DN	Distinguished Name	The full path of an object in the directory
DNS	Domain Name System	Translates names into IP addresses
GLPI	Gestionnaire Libre de Parc Informatique	Open-source ITSM and asset management tool
GPO	Group Policy Object	Settings pushed to the objects in an OU
GUID	Globally Unique Identifier	A permanent unique ID (objectGUID)
ITIL	Information Technology Infrastructure Library	IT service-management best practices
ITSM	IT Service Management	Running IT as a service: tickets, assets, SLAs
LDAP / LDAPS	Lightweight Directory Access Protocol (over SSL/TLS)	Protocol to query a directory; ports 389 / 636
MAC	Media Access Control address	The hardware address of a network card
NAT	Network Address Translation	Shares one address to reach the internet
NXDOMAIN	Non-Existent Domain	DNS answer: "this name does not exist"
OU	Organizational Unit	A folder inside AD
PTR	Pointer record	Reverse DNS: IP → name
RDP	Remote Desktop Protocol	Windows remote desktop, port 3389
SLA	Service Level Agreement	Agreed response / resolution times
SRV	Service record	DNS record that locates a service (e.g. the DCs)
SSH	Secure Shell	Encrypted remote command line, port 22
TLS / SSL	Transport Layer Security / Secure Sockets Layer	Encryption for network traffic
VLAN	Virtual LAN	Splits one physical network into isolated networks
VM	Virtual Machine	A computer emulated in software

Companion to the French lab report in `rapport/`. Real values come from the home lab; no passwords are recorded.